THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS UNIT TEST SET B

Chapters 5–8: **Circles, Measuring Space: Perimeter and Area, The Mathematics of Maybe: Introduction to Probability, Predicting What Comes Next: Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. An arc subtends 28° at a point on the circle. The angle subtended by the same arc at the centre is: 1

(a) 56° **(b)** 152° **(c)** 14° **(d)** 28°

2. The greatest chord of a circle is: 1

(a) the diameter
(b) a chord at distance r from the centre
(c) a chord of length equal to the radius
(d) the radius

3. In a circle of radius 25 cm, a chord is 7 cm from the centre. The length of the chord is: 1

(a) 50 cm **(b)** 24 cm **(c)** 46 cm **(d)** 48 cm

4. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 21 cm is: 1

(a) 1386 cm^2 **(b)** 132 cm^2 **(c)** 693 cm^2 **(d)** 2772 cm^2

5. The area of a triangle with base 16 cm and height 9 cm is: 1

(a) 72 cm^2 **(b)** 88 cm^2 **(c)** 144 cm^2 **(d)** 25 cm^2

6. The length of an arc subtending 45° at the centre of a circle of radius 14 cm ($\pi = \frac{22}{7}$) is: 1

(a) 11 cm **(b)** 22 cm **(c)** 5.5 cm **(d)** 88 cm

7. A letter is picked at random from the word ‘MATHEMATICS’. The probability of picking a vowel is: 1

(a) $\frac{5}{11}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{4}{11}$ **(d)** $\frac{7}{11}$

8. The probability of an impossible event is: 1

(a) 0 **(b)** 0.5 **(c)** −1 **(d)** 1

9. A fair die is rolled once. The probability of getting a number less than 3 is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{3}$

10. For the sequence $t_n = 2 - 5n$, the value of t_6 is: 1

(a) −56 **(b)** −29 **(c)** −27 **(d)** −28

11. Which term of the sequence $t_n = 4n - 3$ is 97? 1

(a) $n = 27$ **(b)** $n = 24$ **(c)** $n = 26$ **(d)** $n = 25$

12. If $s_1 = 3$ and $s_n = s_{n-1}(s_{n-1} - 1)$, then s_3 is: 1

(a) 870 **(b)** 6 **(c)** 30 **(d)** 18

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. ABCD is cyclic with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$. 2

14. The perimeter of a circle is 44 cm. Find its radius ($\pi = \frac{22}{7}$). 2

15. Write the sample space when two coins are tossed together and find the probability of exactly one head. 2

16. A bag has 4 red and 6 blue balls. A ball is drawn at random. What is the probability that it is not red? 2

17. Find the first five terms of $t_n = n^2 - 2n + 3$. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. A chord is 9 cm from the centre of a circle of radius 15 cm. Find its length. 3

19. Two circles of radius r each pass through the centre of the other. Find the perimeter of the shape formed by the outer arcs. 3

20. A die was rolled 120 times: 1→18, 2→22, 3→19, 4→21, 5→24, 6→16. Find the experimental probability of
(i) getting 5
(ii) getting an even number, and compare
(i) with $\frac{1}{6}$. 3

21. Find the 10th term of the GP $3, 6, 12, \ldots$ and its recursive rule. 3

22. Prove that the sum of the opposite angles of a cyclic quadrilateral ABCD is 180° (use the angle-at-centre theorem). 3

OR

Prove that the angle subtended by a diameter at any point on the circle is 90°. 3

23. Find the area of the cyclic quadrilateral with sides 25 cm, 39 cm, 52 cm and 60 cm. 3

OR

Verify Brahmagupta's formula for a square of side a. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. Prove that the angle subtended by an arc at the centre of a circle is double the angle subtended by it at any point on the circle outside the arc (take the case where DC produced meets the circle at E on the arc). 5

OR

Prove that of two chords of a circle, the longer chord is nearer to the centre. 5

25. Rank the following on the probability scale 0 to 1 and label each: 5
(i) The next Monday comes after Sunday.
(ii) It snows in Mumbai in July.
(iii) You roll a 3 on a fair die.
(iv) A fair coin shows heads.
(v) A bag has only red sweets and you draw a red one.

OR

Case study: A well-shuffled pack of 52 cards is used. Find the probability of drawing 5
(i) a king
(ii) a red card
(iii) a face card
(iv) neither a heart nor a diamond
(v) the queen of hearts.

END OF QUESTION PAPER — Total: 50 Marks